K-means clustering

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1. Vector quantization

1.1. Quantizer

Given

- $d, K \in \mathbb{N}^*$
- $\mathcal{C} = \{c_1, \cdots, c_K\}$ a set of K vectors in $\mathbb{R}^d$
- $\mathcal{S} = \{S_1, \cdots, S_K\}$ a partition of $\mathbb{R}^d$ i.e.

$$\mathbb{R}^d = \bigsqcup_{j=1}^{K} S_j$$

A K-quantizer associated with $\mathcal{S}$ and $\mathcal{C}$ is a function $q : \mathbb{R}^d \to \mathbb{R}^d$ such that

$$\forall x \in \mathbb{R}^d,\ \forall j = 1, \cdots, K; \quad x \in S_j \iff q(x) = c_j$$

In such case we call $\mathcal{S}$ the partition associated to q and $\mathcal{C}$ the centers / centroids associated to q.

1.2. Vector Quantization

Let $X \in \mathbb{R}^d$ be a random vector

Vector quantization is the representation of X by $q(X)$ where q is a K-quantizer. q maps vectors of $\mathbb{R}^d$ to a finite subset $\mathcal{C} \implies$ lossy data compression

1.3. Error

Let $X \in \mathbb{R}^d$ be a random vector of distribution μ.

To measure the error of the replacement of X by $q(X)$, one can use:

$$D(\mu, q) = \mathbb{E}\left[\rho(X, q(X))\right] = \int_{x \in \mathbb{R}^d} \rho(x, q(x))\mu(dx)$$

Where ρ is a function that measures the difference/distance between X and $q(X)$

Examples of dissimilarity

Let $x, y \in \mathbb{R}^d$

- Squared Euclidean distance

$$\forall x, y \in \mathbb{R}^d, \quad \rho(x, y) = \|x - y\|^2$$

- Kullback-Leibler divergence

$$\forall x \in \mathbb{R}_+^d,\ \forall y \in (\mathbb{R}_+^*)^d, \quad \rho(x, y) = \sum_{i=1}^{d} x_i \log\left(\frac{x_i}{y_i}\right) - \sum_{i=1}^{d} x_i + \sum_{i=1}^{d} y_i$$

2. Optimal quantizers

In this section we consider the squared Euclidean distance

2.1. Partition condition

Let

- X be a random variable in $\mathbb{R}^d$
- $q : \mathbb{R}^d \to \mathbb{R}^d$ a K-quantizer of X associated to the centers $\mathcal{C} = \{c_1, \cdots, c_K\}$

The K-quantizer q' associated to $\mathcal{C}$ and defined by

$$\forall x \in \mathbb{R}^d,\ q'(x) = \underset{c \in \mathcal{C}}{\operatorname{argmin}} \|x - c\|^2$$

is such that

$$D(\mu, q') \leq D(\mu, q)$$

2.2. Centroid condition

Let

- X be a random variable in $\mathbb{R}^d$
- $q : \mathbb{R}^d \to \mathbb{R}^d$ a K-quantizer of X associated to the partition $\mathcal{S} = \{S_1, \cdots, S_K\}$

The K-quantizer q' associated to the partition $\mathcal{S}$ and associated to centers defined by

$$\forall j = 1, \cdots, K \quad c'_j = \underset{c \in \mathbb{R}^d}{\operatorname{argmin}}\ \mathbb{E}\left[\|X - c\|^2 \mid X \in S_j\right] = \mathbb{E}\left[X \mid X \in S_j\right]$$

is such that

$$D(\mu, q') \leq D(\mu, q)$$

3. Voronoi partition

3.1. Definition

Let $\mathcal{C} = \{c_1, \cdots, c_K\}$ be centroids/centers.

We define the associated Voronoi partition $\mathcal{S} = \{S_1, \cdots, S_K\}$ by :

- $$S_1 = \left\{ x \in \mathbb{R}^d \mid c_1 = \underset{c \in \mathcal{C}}{\operatorname{argmin}} \|x - c\| \right\}$$
- $$\forall j = 2, \cdots, K, \quad S_j = \left\{ x \in \mathbb{R}^d \mid c_j = \underset{c \in \mathcal{C}}{\operatorname{argmin}} \|x - c\| \right\} \setminus \bigsqcup_{l=1}^{j-1} S_l$$

A quantizer q with the centers $\mathcal{C}$ and associated with the Voronoi partition is such that

$$\forall x \in \mathbb{R}^d, \quad \|x - q(x)\|^2 = \min_{c \in \mathcal{C}} \|x - c\|^2$$

Hence the minimization problem of the distortion can be rewritten as a function of $\mathcal{C}$

i.e.

$$D(\mu, \mathcal{C}) = \mathbb{E}\left[\min_{c \in \mathcal{C}} \|X - c\|^2\right]$$

3.2. Theorem

Let X be a random variable such that $\mathbb{E}[\|X\|^2] < +\infty$, and let q be a K-quantizer whose partition $\mathcal{S}$ is the Voronoi partition associated to its centers $\mathcal{C}$ and whose centers satisfy the centroid condition on $\mathcal{S}$ (assuming $\mathbb{P}(X \in S_j) > 0$ for all j).

Then q is a fixed point of Lloyd's algorithm: a single update cannot decrease the distortion, so q is locally optimal. In particular, any globally optimal K-quantizer must satisfy both the partition and centroid conditions; the converse is not true in general.

4. K-means

4.1. Empirical Error

Let $x_1, \cdots, x_n \in \mathbb{R}^d$ be a sample of a random variable X with an unknown distribution μ

To build a K-quantizer of X, since μ is unknown, one can build an **empirical quantizer** by using an empirical estimation of μ such that, for any Borel subset A of $\mathbb{R}^d$

$$\mu_n(A) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \in A\}$$

In such case, $D(\mu_n, q)$ is the empirical estimator of $D(\mu, q) = \mathbb{E}[\rho(X, q(X))]$

$$D(\mu_n, q) = \frac{1}{n} \sum_{i=1}^{n} \rho(x_i, q(x_i)) = \frac{1}{n} \sum_{i=1}^{n} \min_{j=1,\dots,K} \rho(x_i, c_j)$$

Hence for the squared Euclidean distance

$$D(\mu_n, q) = \frac{1}{n} \sum_{i=1}^{n} \min_{j=1,\dots,K} \|x_i - c_j\|^2 := D(\mu_n, \mathcal{C})$$

4.2. Optimal quantizer

Since μ_n is defined from a fixed number of n points, one can prove the existence of an optimal empirical quantizer.

That optimal empirical quantizer is deduced from the Voronoi partition using the previous empirical measure and error.

4.3. Exhaustive method

- Choose a number of classes K
- Find centroids $\mathcal{C} = \{c_1, \dots, c_K\}$ that minimize the empirical distortion

$$D(\mu_n, \mathcal{C}) = \frac{1}{n} \sum_{i=1}^{n} \min_{j=1,\dots,K} \|x_i - c_j\|^2$$

One can try to minimize $D(\mu_n, \mathcal{C})$ by testing all possible assignments of $x_1, \dots, x_n$ into K non-empty unlabeled groups, then computing the corresponding centers $\mathcal{C}$ and keeping the one with the smallest distortion.

The number of possible assignments in K non-empty unlabeled groups is given by the Stirling number of the second kind:

$$S(n, K) = \frac{1}{K!} \sum_{l=1}^{K} (-1)^{K-l} \binom{K}{l} l^n$$

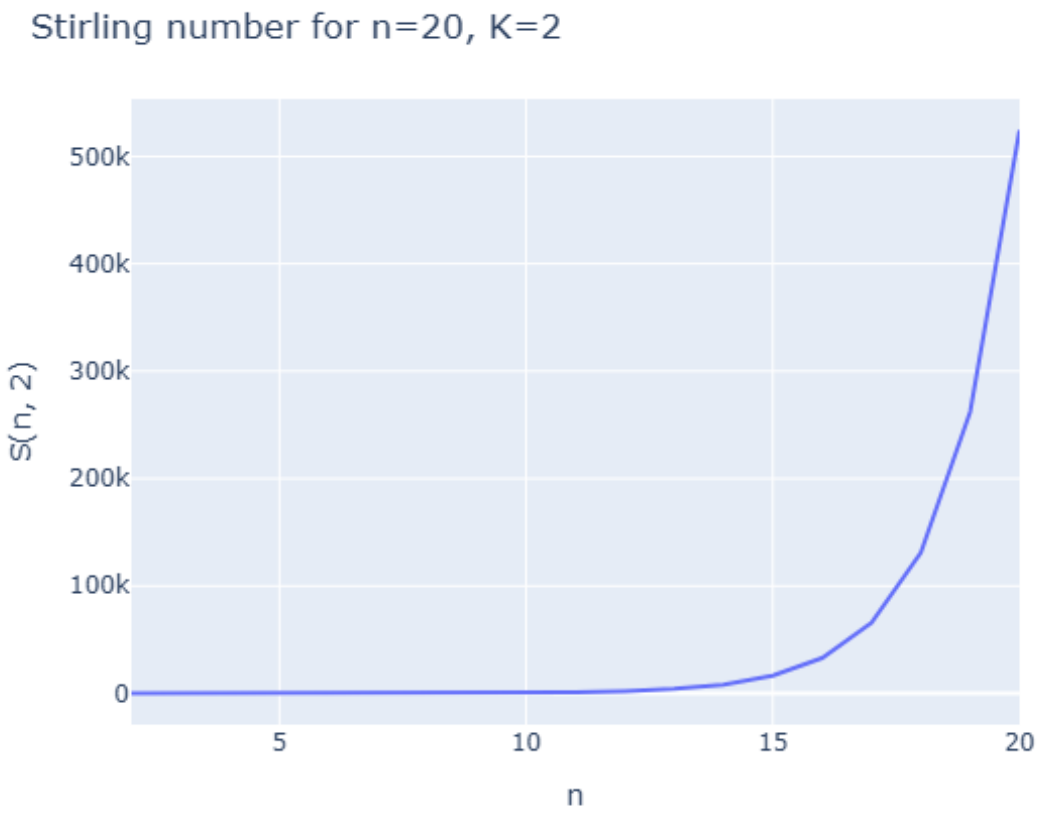


Figure 1: Stirling Number Example|452

Since the number of possible assignments to test becomes very large even for small values of n, K an iterative algorithm is used to approximate the optimal empirical quantizer.

4.4. Lloyd's algorithm

Initialization

- Choose a number of classes K
- Initialize randomly K centers $\mathcal{C}_0 = \{c_{1,0}, \dots, c_{K,0}\}$

Loop

At each step $l = 0, 1, \dots$ until convergence

- $\forall i \in [\![1, n]\!]$, x_i is assigned to the nearest $c_{j,l}$ i.e.

  The partition $\mathcal{S}_l = \{S_{1,l}, \dots, S_{K,l}\}$ is the Voronoi partition associated to $\mathcal{C}_l$ i.e.

$$S_{j,l} = \left\{ x_i \;\middle|\; c_{j,l} = \underset{c \in \mathcal{C}_l}{\operatorname{argmin}}\, \rho(x_i, c), \text{ ties broken by the smallest index} \right\}$$

  Empty clusters are allowed; if $S_{j,l} = \emptyset$, keep the previous center (or reinitialize it) in the update below.

- Update the center using

  $\mathcal{C}_{l+1} = \{c_{1,l+1}, \dots, c_{K,l+1}\}$ such that $\forall j = 1, \dots, K$

$$c_{j,l+1} = \frac{1}{|S_{j,l}|} \sum_{x_i \in S_{j,l}} x_i$$

i.e. the new centers are the means per class, with the empty-cluster convention above

Stopping criterion

When the assignments no longer change (the distortion is non-increasing, so the algorithm terminates)

https://en.wikipedia.org/wiki/Lloyd's_algorithm

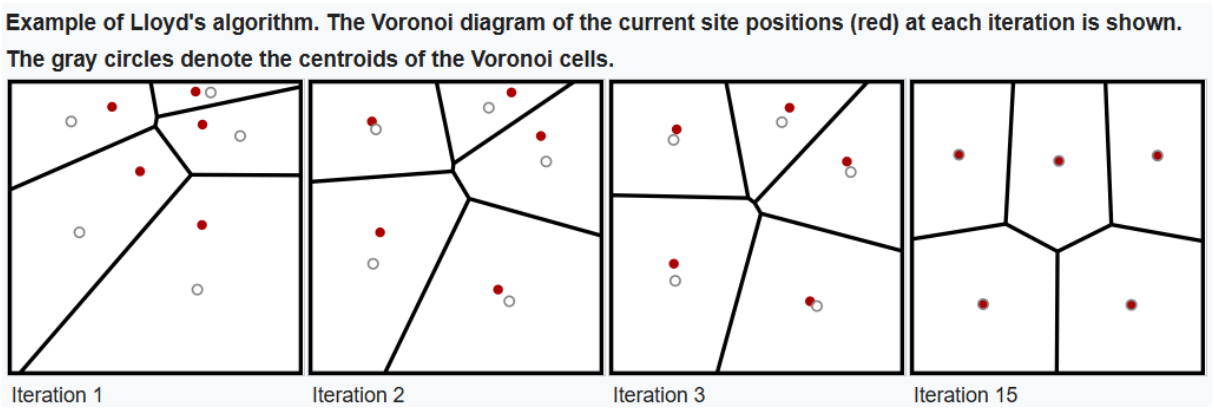


Figure 2: Lloyd's algorithm illustration|452

Remark

This algorithm is quite sensitive to the initialization. It may return only a local minimum.

4.5. Initialization variants

4.5.1. Random initialization

- Run the algorithm with multiple different initializations
- Keep the best result i.e. the one with the least distortion

Initializing from points of the dataset might yield better results

4.5.2. Farthest point

- The first centroid is picked in the sample (uniformly randomly)
- For $j = 2, \dots, K$ the next centroid maximizes the minimum distance to the already chosen centroids

Problem: This method is quite sensitive to outliers!

4.5.3. k-means++

Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be a sample of n points of $\mathbb{R}^d$

In this method we initialized the centers such that

$\mathcal{C}_0 = \{c_{1,0}, \dots, c_{K,0}\}$

$c_{1,0}$ is chosen uniformly, randomly from the sample. For $j = 2, \dots, K$

We associate to each point in the sample: $x_i, i \in [\![1, n]\!]$

For a finite set of centroids $\mathcal{C} \subset \mathbb{R}^d$, we write $\rho(x, \mathcal{C}) := \min_{c \in \mathcal{C}} \rho(x, c)$.

- the squared distances to the nearest previous centroid

$$\rho(x_i, \mathcal{C}_{j-1,0}) = \min_{l \in [\![1, j-1]\!]} \|x_i - c_{l,0}\|^2$$

- the associated probabilities

$$P_j(x_i) = \frac{\rho(x_i, \mathcal{C}_{j-1,0})}{\sum_{k=1}^{n} \rho(x_k, \mathcal{C}_{j-1,0})}$$

Then the next centroid is picked randomly from the sample using the probabilities $P_j(x_i)$

4.6. Convergence

K-means ++

k-means++ leads to a solution close to the optimum. It may be shown that

$$\mathbb{E}\left[D(\mu_n, \mathcal{C}_n^{++})\right] \leq 8(\ln(K) + 2) \min_{\mathcal{C}} D(\mu_n, \mathcal{C})$$

where $\mathcal{C}_n^{++}$ are centroids obtained with k-means++.

4.7. Variants

- K-medoids algorithm
- Other distortion measures
- One can work in more complex spaces than $\mathbb{R}^d$ and classify other objects, including curves, functional data

4.8. Clustering quality metrics

Inertia Let $x_1, \ldots, x_n$ be a sample of n observations and, for each $i \in [\![1, n]\!]$, let $k(i) \in \{1, \ldots, K\}$ be the index of the cluster assigned to x_i and $c_{k(i)}$ its center. One measure of the quality of the clusters is the inertia:

$$J(X, C) = \sum_{i=1}^{n} \|x_i - c_{k(i)}\|^2$$

The lower the inertia, the better.

Silhouette Score The Silhouette score measures how well each point fits its own cluster compared to other clusters. Let x_i be an observation: - a_i = average distance to all other points in its own cluster (cohesion) - b_i = average distance to all points in the nearest other cluster (separation)

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

Therefore $s_i \in [-1, 1]$, and - s_i close to $+1$ means the observation is well inside its cluster and far from other clusters - s_i close to 0 means the observation is on/near a cluster boundary - s_i close to -1 means the observation is closer to another cluster (likely misassigned)

The overall silhouette score is the mean of s_i over all samples. Therefore the higher, the better. However its computation costs $O(n^2)$ distance computations.

4.9. Gaussian mixture sample

On the example below we can see that for the right number of clusters (n_clusters = 3) the inertia is quite low. However, we can also see that for more clusters than necessary (n_clusters > 3) this indicator stays low as well: inertia always decreases as the number of clusters grows, reaching zero when n_clusters = n. Therefore this metric alone is not enough to decide the right number of clusters.

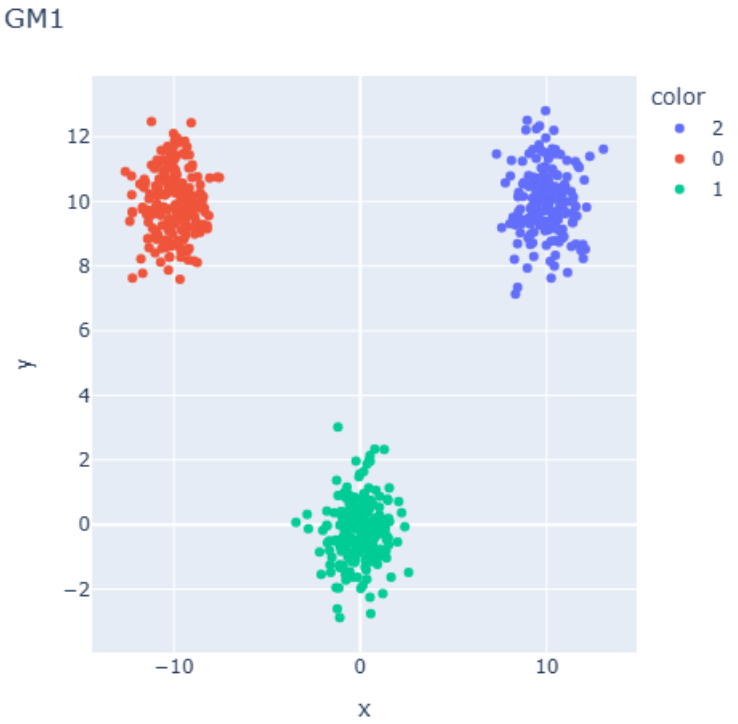


Figure 3: Gaussian mixture dataset|326

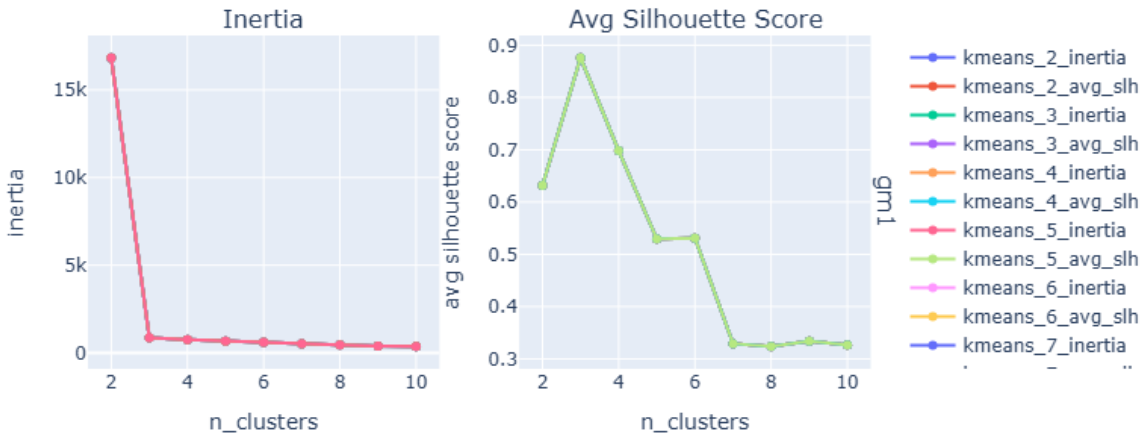


Figure 4: Inertia and average silhouette score vs number of clusters|615

Beyond n_clusters = 3, the inertia does not decrease much, and the average silhouette score is maximal

for n_clusters = 3. Hence one can choose the optimal value n_clusters = 3, which is coherent with the expected result.

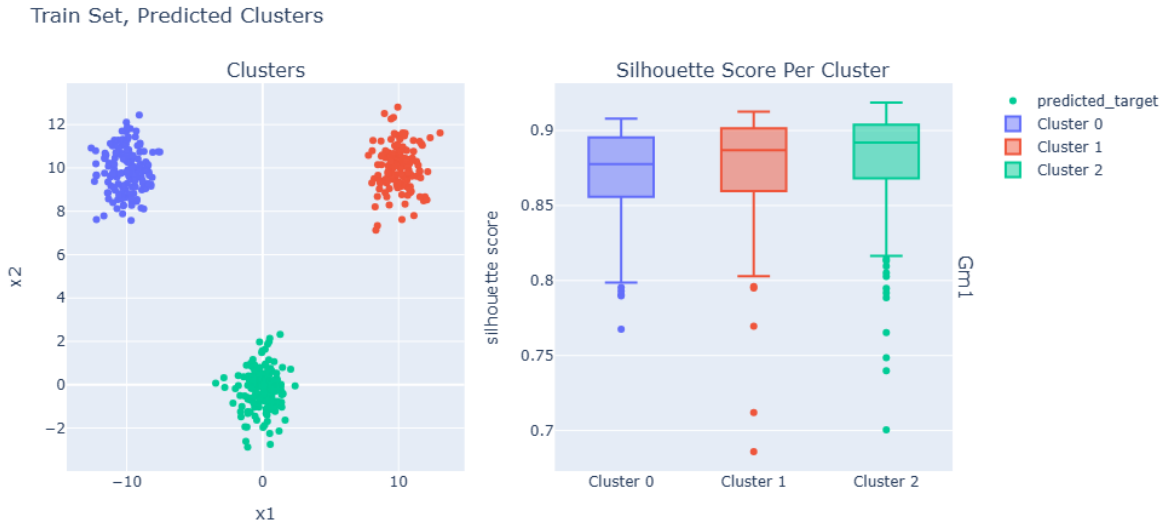


Figure 5: Predicted clusters

4.10. Limits

Example 1: Circles with the same center

In the dataset below, there are 3 clusters, corresponding to the 3 circles. All of them have the same center. We can see that, as expected, the algorithm failed to identify the clusters correctly.



Figure 6: Circles clustering



Figure 7: Circles inertia

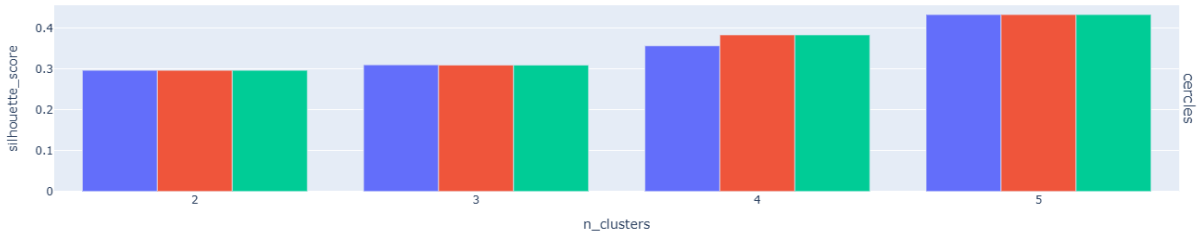


Figure 8: Circles silhouette score

Example 2

In this dataset, the correct clustering is not obvious.

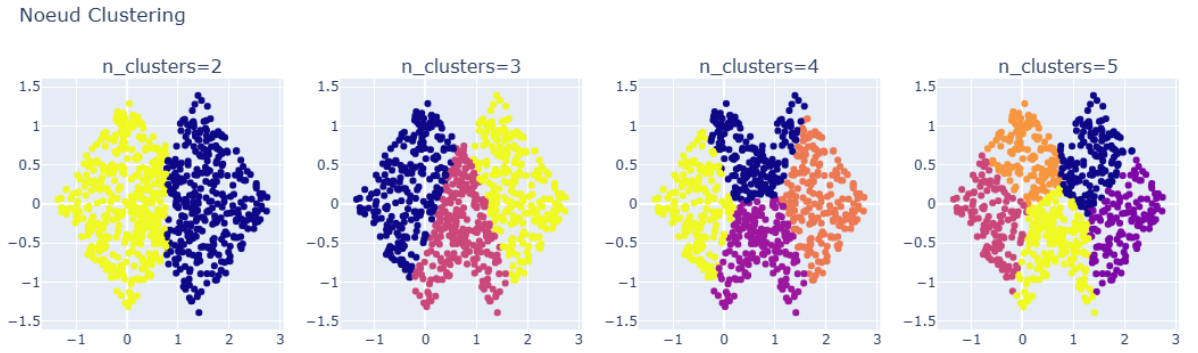


Figure 9: Nodes clustering

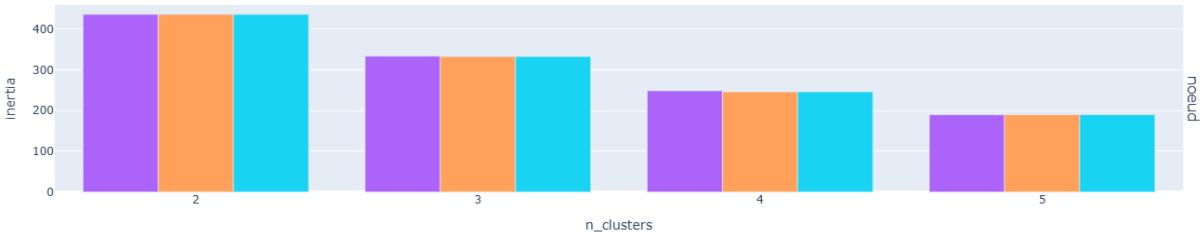


Figure 10: Nodes inertia

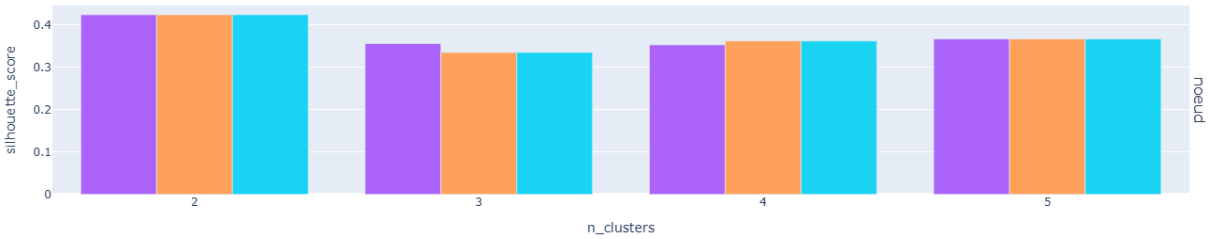


Figure 11: Nodes silhouette score

Although inertia decreases as n_clusters increases, a silhouette score below 0.5 indicates poor performance on this dataset.